

All About FPGAs

A field-programmable gate array (FPGA) is an IC that can be programmed by the user to perform any logic function. As new technologies continue to gain momentum, and the worldwide FPGA market keeps growing, we have some useful websites to help you learn and understand this emerging technology

xilinx.com

Xilinx is a provider of all programmable FPGAs, system on chips (SoCs) and 3D ICs. These devices are coupled with the next-generation design environment and IP to serve a broad range of customer needs, from programmable logic to programmable systems integration. The website has a dedicated section on FPGA tutorials, ISE design suite, ChipScope Pro and embedded development kit.



<http://www.xilinx.com/training/fpga-tutorials.htm>



ni.com

Until now, FPGA technology was available to only engineers with a deep understanding of digital hardware design. But through this site, you can unfold all the details of FPGAs. It allows you to learn about FPGA design with NI LabVIEW system design software. The white paper on the site is a good resource on FPGA fundamentals. The videos on 'Getting Started With LabVIEW FPGA' will surely attract everyone in understanding FPGAs.

<http://www.ni.com/white-paper/5983/en/>

FPGA4Fun

FPGA is a wonderful technology but can be intimidating. This is what this website believes. As many FPGA vendors are providing affordable parts and free tools, this is the right time to learn it. FPGA4fun provides you information on FPGAs and FPGA projects to build with an FPGA board. It provides great examples and a great learning experience for beginners as well as professionals.



<http://www.fpga4fun.com/FPGAinfo1.html>



numato.com

Learning FPGA can be a bit challenging sometimes, but it is a real fun if you have a little patience to read, understand and experiment. This is a site that has an excellent tutorial on FPGAs. This four-part tutorial is not meant to be an in-depth study about Verilog or FPGAs, but it is just a guide to walk you through basic things to design a simple digital circuit in Verilog, simulate it and implement it on hardware.

<http://numato.com/learning-fpga-and-verilog-a-beginners-guide-part-1-introduction>

altera.com

Altera Corporation is a pioneer in programmable logic solutions. It offers FPGAs, SoCs with embedded processor systems, complex programmable logic devices (CPLDs) in combination with software tools, intellectual property, embedded processors and customer support. Apart from several tutorials and online courses, the site has a free online course on 'How to Begin a Simple FPGA Design.' It is meant for engineers who have never designed an FPGA before. Here you will learn about the basic benefits of designing with FPGAs and how to create a simple FPGA design using the Quartus II.



<http://www.altera.com/education/training/courses/ODSW1010>